

Manoj Herath

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Education

2023 – 2026

Ph.D. in Computer Science, Télécom SudParis, Institut Polytechnique de Paris, France (Completion expected in April 2026)

Thesis title: Software Architecture and Modeling of the Digital Twin

Supervisors: Prof. Noel Crespi & Prof. Roberto Minerva, Télécom SudParis, Institut Polytechnique de Paris, France

2016 – 2022

M.Sc. Eng. in Communications and Information Engineering, University of Peradeniya, Sri Lanka (GPA 3.71 out of 4, First Class Honours)

Thesis title: Classification of Hyperspectral Remote Sensing Images with Machine Learning

2009 – 2013

B.Sc. Eng. in Electrical and Electronic Engineering, University of Peradeniya, Sri Lanka (GPA 3.75 out of 4, First Class Honours)

Thesis title: Image-based Sign Language Recognition System for Sinhala Sign Language

Research Interests

- Digital twin modeling and software frameworks
- Communication networks
- IoT sensor network management
- Smart cities
- Edge-cloud continuum resource management
- Model-based systems engineering

Work Experience

2014 – 2022

Engineer, Sri Lanka Telecom, Sri Lanka

Roles: Managing an optical transport network node

Lecturer, Sri Lanka Telecom affiliated with University of Hertfordshire UK

Courses: Digital Signal Processing, Electrical and Electronic Theory, Power Systems

2013 – 2014

Lecturer and Instructor, Department of Electrical and Electronic Engineering, Faculty of Engineering University of Peradeniya

Courses: Power Electronics

2012 – 2013

Trainee Engineer, Sri Lanka Telecom Mobitel, Sri Lanka

Roles: Maintenance of base transceiver stations

Projects

1. 5G Metaverse 2024 – Present

Objective: To develop a digital twin offering real-time collaboration between users, combining virtual and augmented reality.

Contribution: Defining the platform architecture for digital twin use cases, defining the platform APIs, and provisioning the Metaverse digital twin platform.

2. Digital Twin for Cloud Continuum (C2JN) 2023 – 2025
Objective: To set up intelligent spaces associated with business or industrial requests on a sovereign Cloud Continuum.
Contribution: Designing and implementing a comprehensive software architecture that incorporates all essential modules and tools required to realize both smart city and network digital twins. Development and integration of a smart city digital twin for the city of Issy-les-Moulineaux and a network digital twin for the edge–cloud continuum.
3. Situation-Aware Linked heterogeneous Enriched Data (SALTED) 2023 – 2024
Objective: To enrich smart city data for effective management of traffic and urban planning
Contribution: Implementing the injection chain that processes and organizes sensor data collected from Madrid City's vehicular traffic for external applications.

Skills

- Programming: Python, C
- Data & AI: ML pipelines, data workflows
- Cloud & DevOps: Kubernetes, Docker, CI/CD, FastAPI, Postman, Tableau
- Modeling & Simulation: SysML, Eclipse Papyrus, Matlab/Simulink, NGS-LD
- Project Management: Budget planning, scheduling, team coordination
- Operating Systems: Linux, Windows, macOS
- Languages: English (Fluent), French (Basic)
- Academic Skills: Research, teaching, training, supervision

Achievements and Certification

- Chartered Engineer in Electronics & Telecommunications, The Institution of Engineers Sri Lanka
- Medal for Excellence in Lecturer Category, City & Guilds, UK.
- Associate Fellow, Higher Education Academy, UK.
- Next Generation Network Technologies and Future Trends, CETTM, India.
- Certified Trainer in Advanced Diploma in Teaching, Training, and City & Guilds, UK

Research Publication

- I. Herath, M., Alvi, M., Minerva, R., Dutta, H., Crespi, N., & Raza, S. M., (2024). *Smart City Digital Twins : A Modular and Adaptive Architecture for Real-Time Data-Driven Urban Management*, International Conference on Network and Service Management (CNSM).
- II. Raza, S. M., Minerva, R., Crespi, N., Alvi, M., Herath, M., & Dutta, H. (2024). *Exploiting the Efficient Data Modeling in Network Digital Twin to Empower Edge-Cloud Continuum*. International Conference on Network and Service Management (CNSM).
- III. Raza, S. M., Minerva, R., Crespi, N., Alvi, M., Herath, M., & Dutta, H. (2025). *A comprehensive survey of Network Digital Twin architecture, capabilities, challenges, and requirements for Edge-Cloud Continuum*, Computer Communications.
- IV. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *An AI-driven, scalable, and modular digital twin framework for traffic management*, International Conference on Wireless Communications and Networking Conference (WCNC).
- V. Alvi, M., Dutta, H., Minerva, R., Crespi, N., Raza, S. M., & Herath, M. (2025). *Global perspectives on digital twin smart cities: Innovations, challenges, and pathways to a sustainable urban future*, Sustainable Cities and Society.
- VI. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *Smart City Digital Twin Edge-Core Deployment: A Case Study on Traffic and Air Quality Management*, International Conference on Network Softwarization (NetSoft).
- VII. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *An Integrated Digital Twin Architecture for Real-Time Urban Air Quality Management*, International Conference on Dependable Systems and Networks (DSN).

Under Review Papers

- I. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *Edge-Cloud Continuum Deployment of Digital Twins for Smart City Metaverse: from Software Modeling to Hardware Implementation*, IEEE Internet of Things.

- II. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *Smart City Digital Twin with Distributed Edge Intelligence and Real-Time Decision Support*, IEEE Internet of Things Magazine.
- III. Rajapaksha P., Herath M., Minerva, R., & Crespi, N. (2025). *Intelligent Digital Twins: Leveraging AI for Modeling, Simulation, and Decision-Making in Complex Systems*, ACM Transactions on Autonomous and Adaptive Systems.
- IV. Kannappan R., Herath, M., Julien H., Emmanuel B., & Crespi N. (2025). *Decentralized Digital Twin Ecosystems: Selective Data Disclosure of Heterogeneous Twins for Digital Product Passport*, International Conference in Pervasive Computing.
- V. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *Distributed Implementation of Smart City Digital Twin with Federated Learning-assisted Prediction*, ACM Transactions on Cyber Physical Systems.
- VI. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *MBSE-Driven Behavior Modeling of Industrial Digital Twins: Methodology and Case Study*, Information and Software Technology.
- VII. Alvi, M., Dutta, H., Minerva, R., Crespi, N., Raza, S. M., & Herath, M. (2025). *A Predictive Digital Twin Framework for Real-time Urban Traffic Management*, IEEE Internet of Things Journal.
- VIII. Herath, M., Dutta, H., Alvi, M., Minerva, R., Crespi, N., & Raza, S. M., (2025). *Federated LSTM assisted Digital Twin for Multimodal Heterogeneous Data in Smart City Metaverse*, Network Operations and Management Symposium.

Reviewer Role

International Conference on 6G Networking, IEEE Journals

Reference

Noel Crespi, Professor, Télécom SudParis, Institut Polytechnique de Paris
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Roberto Minerva, Professor, Télécom SudParis, Institut Polytechnique de Paris
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